

Climate 5000 Series - Gen 4

18K BTU Multi Zone Max Performance Outdoor Unit

Engineering
Submittal
Sheet



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Overview and Project Information



Project name

Location

Architect

Engineer

General contractor

Reference #

Date

18K Btu Multi Zone Max Performance Outdoor Unit

Outdoor part number

8-733-962-699

Outdoor model number

BMS500-AAM018-1CSXHD

Efficiency Data

Ducted

Non-Ducted

Mixed Ducted & Non-Ducted

SEER2

20

25

22.5

EER2

13

15

14.0

HSPF2 (Region IV)

10

10.4

10.2

HSPF2 (Region V)

8

8.5

8.1

COP(47°F)

4

4

4.0

COP (5°F)

2.0

2.0

2.0

Performance Data

Ducted

Non-Ducted

Mixed Ducted & Non-Ducted

Cooling

Capacity Range (Btu/h)

5400~27600

5400~28000

5400-27800

Rated Capacity (Btu/h)

19000

19000

19000

Current (A)

7.5

5.6

8.1

Input Power (W)

1460

1265

1794.5

Heating

Capacity Range (Btu/h)

5700~25400

5700~27000

5700-26200

Rated Capacity (47°F)

20000

20000

20000

Capacity (5°F)

18000

19000

18850

Current (A)

7.5

6.5

8.3

Input Power (W)

1465

1465

1872

Operation Range

ODU	Cooling (°F / °C)	-22~122 / -30~50
	Heating (°F / °C)	-22~75 / -30~24

Electrical Data

Power supply (V / Ø / Hz)	208-230 / 1 / 60
Min. circuit ampacity (A)	17.5
Max. fuse (A)	20

Compressor Data

Type:	Rotary
Refrigerant oil /oil charge (ml)	VG74 / 620
Input (W)	805 / 2045
Rated current (A)	4.15 / 9.30

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Piping	
Refrigerant precharge (ft / m)	24.6(7.5)x3
Refrigerant charge (oz / kg)	78.31 / 2.22
Liquid side (in / mm)	3x1/4in / 3x6.35mm
Gas side (in / mm)	3x3/8in / 3x9.52mm
Additional Charge of Refrigerant - liquid side (oz/ft)/(g/m)	0.16 / 15
Additional Charge of Refrigerant - gas side (oz/ft)/(g/m)	0.32 / 30
Max. refrigerant pipe length (ft / m)	196.8 / 60
Max. length to one IDU (ft / m)	98.4 / 30
Max. height difference between IDU and ODU (ft / m)	49.2 / 15
Max. height difference between IDU's (ft / m)	33 / 10

System Data		
Refrigerant type		R454B
Design pressure (PSIG)		550 / 340
Dehumidification (L/hr)		0
ODU	Fan motor model #	ZKFN-120-8-9
	Fan motor qty (ea)	1
	Fan motor input (W)	150.0
	Fan motor speed, H / M / L (RPM)	800 / 700 / 650
	Air flow, max (CFM)	2130.7
	Noise level, max (dB(A))	61.5
	Net weight (lbs / kg)	137.35 / 62.3
	Gross weight (lbs / kg)	148.15 / 67.2
IDU	Minimum # of indoors	1
	Maximum # of indoors	3

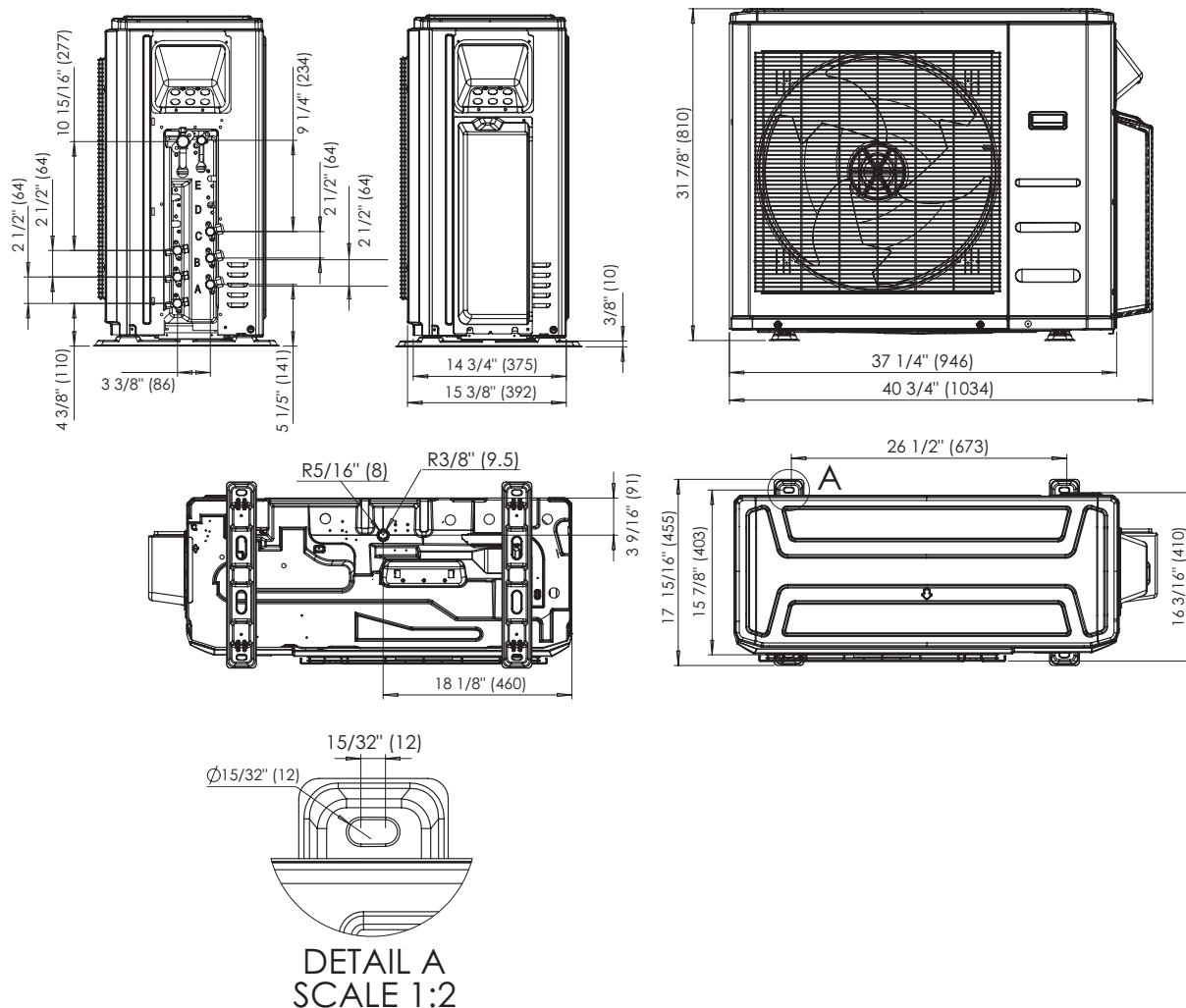
Multi Zone Pairing Information	1 ZONE	2 ZONE	3 ZONE		4 ZONE		5 ZONE
BMS500-AAM018-1CSXHD	18	6+6	6+6+6				
	24	6+9	6+6+9				
		6+12	6+6+12				
		6+18	6+6+18				
		6+24	6+6+24				
		9+9	6+9+9				
		9+12	6+9+12				
		9+18	6+9+18				
		9+24	6+12+12				
		12+12	6+12+18				
		12+18	9+9+9				
		12+24	9+9+12				
		18+18	9+9+18				
			9+12+12				
			12+12+12				
Capacities	Wall Mounted - 6K to 24K Cassette/Ducted - 9K to 24K						

18K Btu Multi Zone Max Performance Outdoor Unit



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Outdoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)
BMS500-AAM018-1CSXHD	37.24	16.14	31.89

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18K Btu Multi Zone Max Performance Outdoor Unit


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Cooling Performance - Indoor Dry Bulb Temperature at 971CFM (Ducted)

ODU DB(°F)	IDU WB (°F)	60.8°F (16.0°C)					64.4°F (18.0°C)					67°F (19.4°C)					71.6°F (22.0°C)				
	IDU DB (°F)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)
-22°F (-30.0°C)	TC	22512	22542	22566	22567	22598	23868	23896	23920	23925	23953	25294	25327	25355	25360	25388	26818	26847	26877	26881	26914
	S/T	0.60	0.71	0.80	0.80	0.91	0.57	0.67	0.76	0.76	0.86	0.53	0.63	0.71	0.71	0.81	0.50	0.59	0.67	0.67	0.77
	PI	1.69	1.69	1.69	1.69	1.70	1.79	1.79	1.80	1.80	1.80	1.90	1.90	1.90	1.90	1.91	2.01	2.02	2.02	2.02	2.02
-13°F (-25.0°C)	TC	24387	24420	24447	24448	24481	25857	25888	25913	25919	25949	27402	27438	27468	27473	27504	29053	29085	29116	29121	29157
	S/T	0.61	0.72	0.82	0.82	0.93	0.58	0.68	0.77	0.77	0.88	0.54	0.64	0.73	0.73	0.83	0.51	0.60	0.68	0.68	0.78
	PI	1.75	1.76	1.76	1.76	1.76	1.86	1.86	1.86	1.86	1.87	1.97	1.97	1.97	1.98	1.98	2.09	2.09	2.09	2.09	2.10
-4°F (-20.0°C)	TC	26263	26299	26327	26328	26364	27846	27879	27907	27912	27945	29510	29548	29581	29587	29620	31288	31322	31356	31361	31400
	S/T	0.62	0.73	0.83	0.83	0.94	0.59	0.69	0.78	0.78	0.89	0.55	0.65	0.74	0.74	0.84	0.52	0.61	0.70	0.70	0.79
	PI	1.82	1.82	1.82	1.82	1.82	1.93	1.93	1.93	1.93	1.93	2.04	2.04	2.05	2.05	2.05	2.16	2.17	2.17	2.17	2.17
0°F (-17.8°C)	TC	27097	27134	27163	27164	27201	28730	28764	28793	28798	28833	30447	30486	30520	30526	30560	32281	32316	32351	32356	32397
	S/T	0.63	0.73	0.84	0.84	0.95	0.59	0.69	0.79	0.79	0.90	0.56	0.66	0.74	0.74	0.85	0.53	0.62	0.70	0.70	0.80
	PI	1.84	1.85	1.85	1.85	1.85	1.96	1.96	1.96	1.96	1.96	2.07	2.07	2.08	2.08	2.08	2.20	2.20	2.20	2.20	2.20
5°F (-15.0°C)	TC	28139	28177	28208	28209	28247	29835	29871	29900	29906	29941	31618	31659	31694	31700	31735	33523	33559	33596	33601	33643
	S/T	0.63	0.74	0.84	0.84	0.96	0.60	0.70	0.80	0.80	0.91	0.56	0.66	0.75	0.75	0.86	0.53	0.62	0.71	0.71	0.81
	PI	1.88	1.88	1.88	1.88	1.89	1.99	1.99	2.00	2.00	2.00	2.11	2.11	2.12	2.12	2.12	2.24	2.24	2.24	2.24	2.25
14°F (-10.0°C)	TC	28530	28569	28599	28601	28639	30250	30285	30315	30321	30357	32057	32098	32134	32140	32176	33988	34025	34062	34068	34110
	S/T	0.62	0.73	0.83	0.83	0.94	0.59	0.69	0.78	0.78	0.89	0.55	0.65	0.74	0.74	0.84	0.52	0.61	0.69	0.69	0.79
	PI	1.85	1.85	1.85	1.85	1.86	1.96	1.96	1.97	1.97	1.97	2.08	2.08	2.08	2.08	2.09	2.20	2.21	2.21	2.21	2.21
17°F (-8.3°C)	TC	28660	28699	28730	28731	28770	30388	30424	30454	30460	30496	32203	32245	32281	32287	32323	34143	34181	34218	34223	34266
	S/T	0.62	0.72	0.82	0.82	0.94	0.58	0.68	0.78	0.78	0.88	0.55	0.64	0.73	0.73	0.83	0.52	0.61	0.69	0.69	0.79
	PI	1.84	1.84	1.84	1.85	1.85	1.95	1.95	1.96	1.96	1.96	2.07	2.07	2.07	2.07	2.08	2.19	2.19	2.20	2.20	2.20
23°F (-5.0°C)	TC	29431	29470	29502	29504	29543	31205	31242	31272	31279	31316	33069	33112	33149	33155	33192	35061	35099	35138	35143	35187
	S/T	0.61	0.72	0.81	0.81	0.93	0.58	0.68	0.77	0.77	0.88	0.54	0.64	0.72	0.72	0.83	0.51	0.60	0.68	0.68	0.78
	PI	1.89	1.89	1.89	1.89	1.90	2.00	2.00	2.01	2.01	2.01	2.12	2.12	2.13	2.13	2.13	2.25	2.25	2.25	2.25	2.26
35°F (1.7°C)	TC	30972	31013	31047	31048	31090	32838	32877	32910	32916	32955	34800	34845	34884	34891	34930	36897	36937	36977	36983	37029
	S/T	0.60	0.70	0.80	0.80	0.91	0.56	0.66	0.75	0.75	0.86	0.53	0.62	0.71	0.71	0.81	0.50	0.59	0.67	0.67	0.76
	PI	1.98	1.99	1.99	1.99	1.99	2.10	2.11	2.11	2.11	2.11	2.23	2.23	2.24	2.24	2.24	2.36	2.37	2.37	2.37	2.37
41°F (5.0°C)	TC	29981	30022	30054	30055	30096	31788	31826	31857	31863	31901	33687	33731	33769	33775	33813	35717	35756	35795	35800	35845
	S/T	0.61	0.72	0.82	0.82	0.93	0.58	0.68	0.77	0.77	0.88	0.54	0.64	0.73	0.73	0.83	0.51	0.60	0.69	0.69	0.78
	PI	2.01	2.02	2.02	2.02	2.02	2.13	2.14	2.14	2.14	2.14	2.26	2.26	2.27	2.27	2.27	2.40	2.40	2.40	2.40	2.41
47°F (8.3°C)	TC	28991	29030	29061	29062	29101	30738	30774	30805	30811	30847	32574	32616	32653	32659	32695	34537	34574	34612	34617	34661
	S/T	0.63	0.73	0.83	0.83	0.95	0.59	0.69	0.79	0.79	0.90	0.56	0.65	0.74	0.74	0.85	0.53	0.62	0.70	0.70	0.80
	PI	2.04	2.04	2.05	2.05	2.05	2.16	2.17	2.17	2.17	2.17	2.29	2.30	2.30	2.30	2.30	2.43	2.43	2.44	2.44	2.44
50°F (10.0°C)	TC	28488	28526	28557	28558	28596	30204	30240	30270	30276	30312	32009	32051	32086	32092	32128	33938	33974	34011	34017	34059
	S/T	0.62	0.73	0.82	0.82	0.94	0.58	0.68	0.78	0.78	0.89	0.55	0.65	0.73	0.73	0.84	0.52	0.61	0.69	0.69	0.79
	PI	2.02	2.02	2.03	2.03	2.03	2.14	2.15	2.15	2.15	2.15	2.27	2.27	2.28	2.28	2.28	2.41	2.41	2.41	2.41	2.42
59°F (15.0°C)	TC	26978	27015	27044	27045	27081	28604	28638	28666	28672	28706	30313	30353	30386	30392	30426	32140	32175	32210	32215	32255
	S/T	0.60	0.70	0.80	0.80	0.91	0.56	0.66	0.75	0.75	0.86	0.53	0.62	0.71	0.71	0.81	0.50	0.59	0.67	0.67	0.76
	PI	1.96	1.97	1.97	1.97	1.97	2.08	2.08	2.09	2.09	2.09	2.21	2.21	2.21	2.21	2.21	2.34	2.34	2.34	2.34	2.35
68°F (20.0°C)	TC	25469	25504	25531	25532	25567	27004	27036	27063	27068	27100	28618	28655	28687	28692	28724	30342	30375	30408	30413	30450
	S/T	0.58	0.68	0.77	0.77	0.88	0.54	0.64	0.72	0.72	0.83	0.51	0.60	0.68	0.68	0.78	0.48	0.57	0.65	0.65	0.74
	PI	1.90	1.91	1.91	1.91	1.91	2.02	2.02	2.02	2.02	2.03	2.14	2.14	2.14	2.15	2.15	2.27	2.27	2.27	2.27	2.28
77°F (25.0°C)	TC	23960	23992	24018	24019	24052	25404	25434	25459	25464	25495	26922	26957	26987	26992	27022	28544	28575	28606	28610	28646
	S/T	0.55	0.65	0.74	0.74	0.84	0.52	0.61	0.70	0.70	0.80	0.49	0.58	0.66	0.66	0.75	0.47	0.55	0.62	0.62	0.71
	PI	1.85	1.85	1.85	1.85	1.85	1.96	1.96	1.96	1.96	1.96	2.07	2.08	2.08	2.08	2.08	2.20	2.20	2.20	2.20	2.21
82°F (27.8°C)	TC	23753	23785	23811	23812	23844	25185	25215	25240	25245	25275	26690	26724	26754	26759	26789	28298	28328	28359	28364	28399
	S/T	0.56	0.66	0.75	0.75	0.85	0.53	0.62	0.70	0.70	0.80	0.50	0.58	0.66	0.66	0.76	0.47	0.55	0.63	0.63	0.71
	PI	1.87	1.87	1.87	1.87	1.88	1.98	1.98	1.99	1.99	1.99	2.10	2.10	2.10	2.11	2.11	2.23	2.23	2.23	2.23	2.23
86°F (30.0°C)	TC	24385	24418	24444	24445	24478	25855	25885	25911	25916	25947	27399	27435	27466	27471	27501	29050	29082	29114	29118	29154
	S/T	0.58	0.69	0.78	0.78	0.89	0.55	0.65	0.74	0.74	0.84	0.52	0.61	0.69	0.69	0.79	0.49	0.58	0.65	0.65	0.75
	PI	1.96	1.96	1.96	1.96	1.96	2.07	2.08	2.08	2.08	2.08	2.20	2.20	2.20	2.20	2.21	2.33	2.33	2.34	2.34	2.34
95°F (35.0°C)	TC	25807	25842	25869	25871	25905	27362	27394	27422	27427	27460	28997	29034	29067	29072	29105	30744	30777	30811	30816	30854
	S/T	0.64	0.75	0.85	0.85	0.97	0.60	0.71	0.81	0.81	0.92	0.57	0.67	0.76	0.76	0.87	0.54	0.63	0.72	0.72	0.82
	PI	2.15	2.16	2.16	2.16	2.16	2.28	2.29	2.29	2.29	2.29	2.42	2.42	2.43	2.43	2.43	2.57	2.57	2.57	2.57	2.58
104°F (40.0°C)	TC	23080																			

18K Btu Multi Zone Max Performance Outdoor Unit



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Cooling Performance - Indoor Dry Bulb Temperature at 1147 CFM (Non-Ducted)																					
ODU DB(F)	IDU WB (F)	60.8°F(16.0°C)					64.4°F(18.0°C)					67°F(19.4°C)					71.6°F(22.0°C)				
	IDU DB (F)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)
-22°F (-30.0°C)	TC	21380	21409	21432	21433	21461	22668	22695	22717	22722	22749	24023	24054	24081	24085	24112	25470	25498	25525	25529	25561
	S/T	0.60	0.71	0.80	0.80	0.91	0.57	0.67	0.76	0.76	0.86	0.53	0.63	0.71	0.71	0.81	0.50	0.59	0.67	0.67	0.77
	PI	1.94	1.95	1.95	1.95	1.95	2.06	2.06	2.07	2.07	2.07	2.19	2.19	2.19	2.19	2.19	2.32	2.32	2.32	2.32	2.33
-13°F (-25.0°C)	TC	23161	23193	23218	23219	23250	24557	24586	24611	24615	24645	26024	26058	26087	26092	26121	27592	27622	27652	27657	27691
	S/T	0.61	0.72	0.82	0.82	0.93	0.58	0.68	0.77	0.77	0.88	0.54	0.64	0.73	0.73	0.83	0.51	0.60	0.68	0.68	0.78
	PI	2.02	2.02	2.02	2.02	2.02	2.14	2.14	2.14	2.14	2.15	2.27	2.27	2.27	2.27	2.27	2.40	2.41	2.41	2.41	2.41
-4°F (-20.0°C)	TC	24943	24977	25004	25005	25038	26446	26478	26504	26509	26540	28026	28063	28094	28099	28131	29715	29747	29780	29784	29821
	S/T	0.62	0.73	0.83	0.83	0.94	0.59	0.69	0.78	0.78	0.89	0.55	0.65	0.74	0.74	0.84	0.52	0.61	0.70	0.70	0.79
	PI	2.09	2.09	2.09	2.09	2.10	2.21	2.22	2.22	2.22	2.22	2.35	2.35	2.35	2.35	2.36	2.49	2.49	2.49	2.49	2.50
0°F (-17.8°C)	TC	25735	25770	25797	25798	25833	27286	27318	27345	27351	27383	28916	28954	28986	28991	29024	30658	30692	30725	30730	30768
	S/T	0.63	0.73	0.84	0.84	0.95	0.59	0.69	0.79	0.79	0.90	0.56	0.65	0.74	0.74	0.85	0.53	0.62	0.70	0.70	0.80
	PI	2.12	2.12	2.13	2.13	2.13	2.25	2.25	2.25	2.25	2.26	2.38	2.39	2.39	2.39	2.39	2.53	2.53	2.53	2.53	2.54
5°F (-15.0°C)	TC	26725	26761	26789	26791	26827	28335	28369	28397	28402	28436	30028	30067	30101	30106	30140	31837	31872	31907	31912	31951
	S/T	0.63	0.74	0.84	0.84	0.96	0.60	0.70	0.80	0.80	0.91	0.56	0.66	0.75	0.75	0.86	0.53	0.62	0.71	0.71	0.81
	PI	2.16	2.16	2.17	2.17	2.17	2.29	2.29	2.30	2.30	2.30	2.43	2.43	2.43	2.43	2.44	2.57	2.58	2.58	2.58	2.58
14°F (-10.0°C)	TC	25777	25812	25840	25841	25876	27331	27363	27390	27396	27428	28964	29001	29034	29039	29071	30709	30742	30776	30780	30819
	S/T	0.62	0.73	0.83	0.83	0.94	0.59	0.69	0.78	0.78	0.89	0.55	0.65	0.74	0.74	0.84	0.52	0.61	0.69	0.69	0.79
	PI	2.18	2.18	2.18	2.18	2.19	2.31	2.31	2.31	2.31	2.32	2.45	2.45	2.45	2.45	2.46	2.59	2.60	2.60	2.6	

Bolded values were tested. All other values are simulated and for reference only.

Bosch Thermotechnology Corp.
Londonderry, NH • Watertown, MA • Ft. Lauderdale, FL

Climate 5000 Series - Gen 4

18K Btu Multi Zone Max Performance Outdoor Unit



Heating Performance - Indoor Dry Bulb Temperature at 971 CFM (Ducted)										
Outdoor DB(°F)	TCITotal Heating Capacity(BTU/hr)					PIPower Input(kW)				
	Indoor Conditions(DB °F)					Indoor Conditions(DB °F)				
	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)
-22°F(-30.0°C)	11257	11032	10976	10932	10834	2.17	2.19	2.19	2.20	2.21
-13°F(-25.0°C)	14070	13789	13720	13665	13542	2.32	2.34	2.34	2.35	2.36
-4°F(-20.0°C)	16484	16155	16074	16010	15866	2.45	2.47	2.48	2.48	2.49
0°F(-17.8°C)	17470	17121	17035	16967	16815	2.52	2.54	2.55	2.55	2.56
5°F(-15.0°C)	18703	18329	18237	18164	18001	2.60	2.62	2.63	2.64	2.65
14°F(-10.0°C)	22104	21662	21554	21467	21274	2.61	2.63	2.63	2.64	2.65
17°F(-8.3°C)	23238	22773	22659	22568	22365	2.61	2.63	2.63	2.64	2.65
22°F(-5.6°C)	22492	22042	21932	21844	21648	2.47	2.49	2.49	2.50	2.51
27°F(-2.8°C)	21747	21312	21205	21120	20930	2.33	2.35	2.36	2.36	2.37
32°F(-0.0°C)	21001	20581	20478	20396	20213	2.20	2.21	2.22	2.22	2.23
35°F(1.7°C)	20554	20143	20042	19962	19782	2.11	2.13	2.14	2.14	2.15
37°F(2.8°C)	22418	21969	21860	21772	21576	2.14	2.15	2.16	2.16	2.17
42°F(5.6°C)	27077	26536	26403	26298	26061	2.19	2.21	2.21	2.22	2.23
44.6°F(7.0°C)	28941	28362	28221	28108	27855	2.21	2.23	2.24	2.24	2.25
47°F(8.3°C)	31737	31102	30947	30823	30546	2.25	2.27	2.27	2.28	2.28
52°F(11.1°C)	29557	28966	28821	28706	28448	1.95	1.97	1.97	1.97	1.98
57°F(13.9°C)	27378	26830	26696	26589	26350	1.65	1.67	1.67	1.67	1.68
59°F(15.0°C)	26506	25976	25846	25742	25511	1.53	1.55	1.55	1.55	1.56
62°F(16.7°C)	25198	24694	24570	24472	24252	1.36	1.37	1.37	1.37	1.38
64.4°F(18.0°C)	24326	23839	23720	23625	23413	1.24	1.25	1.25	1.25	1.26

Bolded values were tested. All other values are simulated and for reference only.

Climate 5000 Series - Gen 4

18K Btu Multi Zone Max Performance Outdoor Unit



BOSCH

Heating Performance - Indoor Dry Bulb Temperature at 1147 CFM (Non-Ducted)										
Outdoor DB(°F)	TC Total Heating Capacity(BTU/hr)					PI Power Input(kW)				
	Indoor Conditions(DB °F)					Indoor Conditions(DB °F)				
	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)
-22°F(-30.0°C)	10329	10123	10072	10032	9942	2.08	2.09	2.10	2.10	2.11
-13°F(-25.0°C)	12975	12715	12652	12601	12488	2.22	2.24	2.24	2.25	2.26
-4°F(-20.0°C)	16085	15764	15685	15622	15482	2.39	2.41	2.41	2.41	2.42
0°F(-17.8°C)	17697	17343	17256	17187	17032	2.48	2.50	2.51	2.51	2.52
5°F(-15.0°C)	19711	19316	19220	19143	18971	2.60	2.62	2.62	2.63	2.64
14°F(-10.0°C)	20277	19872	19773	19693	19516	2.55	2.57	2.58	2.58	2.59
17°F(-8.3°C)	20466	20057	19957	19877	19698	2.54	2.56	2.56	2.57	2.58
22°F(-5.6°C)	21163	20740	20636	20554	20369	2.46	2.48	2.49	2.49	2.50
27°F(-2.8°C)	21860	21423	21316	21231	21040	2.39	2.41	2.41	2.42	2.43
32°F(0.0°C)	22557	22106	21995	21907	21710	2.31	2.33	2.33	2.34	2.35
35°F(1.7°C)	22975	22516	22403	22314	22113	2.27	2.28	2.29	2.29	2.30
37°F(2.8°C)	23717	23242	23126	23033	22826	2.26	2.27	2.28	2.28	2.29
42°F(5.6°C)	25570	25058	24933	24833	24610	2.23	2.25	2.26	2.26	2.27
44.6°F(7.0°C)	26311	25785	25656	25553	25323	2.22	2.24	2.25	2.25	2.26
47°F(8.3°C)	27423	26874	26740	26633	26393	2.21	2.23	2.23	2.24	2.25
52°F(11.1°C)	25481	24971	24846	24747	24524	1.91	1.92	1.93	1.93	1.94
57°F(13.9°C)	23539	23068	22953	22861	22655	1.60	1.62	1.62	1.62	1.63
59°F(15.0°C)	22762	22307	22195	22106	21907	1.48	1.49	1.50	1.50	1.51
62°F(16.7°C)	21597	21165	21059	20975	20786	1.30	1.31	1.31	1.32	1.32
64.4°F(18.0°C)	20820	20403	20301	20220	20038	1.18	1.19	1.19	1.19	1.20

Bolded values were tested. All other values are simulated and for reference only.